

QUBO: A Retrieval-Augmented Generation powered Chatbot for Supporting Student Learning through Lecture Notes

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I. INTRODUCTION

The massive volume of information students encounter in today's digital age often leads to significant difficulties, including information overload and a lack of effective organization, which can hinder academic progress [1], [2]. Although detailed lecture notes and digital materials are provided, their inherent complexity and length can unfortunately create a high cognitive load, counteracting effective learning [3]. Cognitive Load Theory (CLT) posits that lengthy or poorly organized educational materials strain learners' limited working memory, hindering their ability to process and retain information [4], [5]. Consequently, research shows that students increasingly rely on Artificial Intelligence (AI) powered chatbots for academic help, with studies indicating that two out of three students use these tools weekly to clarify doubts and enhance their learning experience [6]–[8].

However, the reliance on general-purpose Large Language Models (LLMs) introduces critical problems and limitations. These models are susceptible to hallucination, generating responses that appear plausible but are factually incorrect or inconsistent with specific academic sources [10]. Furthermore, their reliance on vast, general training data means the information provided may be outdated or misaligned with a university's specific curriculum and learning goals [9]. This unreliability is a significant concern in educational contexts, where factual accuracy is crucial, potentially leading to student confusion, misinformation, and compromised learning outcomes. With students increasingly relying on these tools, mitigating hallucinations and ensuring alignment with educational content is crucial [11].

A. Statement of the Problem and Motivation

The primary challenge is to develop a trustworthy AI-powered learning assistant that ensures knowledge reliability by grounding responses directly in authoritative educational materials. Retrieval-Augmented Generation (RAG) addresses the hallucination problem inherent in base Large Language Models (LLMs). RAG improves LLM performance by connecting the model to trusted, external knowledge sources. Grounding responses in relevant passages retrieved from lecture notes minimizes hallucination and enables the model to produce accurate and contextually aligned answers. Although RAG demonstrates promising enhancements in educational contexts, there is limited investigation into its real-world effectiveness in supporting student comprehension and learning from university-provided lecture notes.

B. Objectives and Contributions

Qubo, a Retrieval-Augmented Generation (RAG) powered chatbot, is introduced to support student learning by providing accurate and contextually grounded responses sourced directly from lecture notes.

The general objective is to design and develop Qubo as a reliable supplemental learning tool. The specific contributions are as follows:

- 1) **RAG Pipeline Development:** To develop a robust RAG pipeline capable of processing and understanding diverse academic lecture note formats (PDF, DOCX, text) as the primary knowledge source.
- 2) **Effective Retrieval Strategies:** To develop retrieval methods that improve the accuracy and contextual com-

prehension of information extracted from varied lecture note structures.

- 3) **Response Generation Refinement:** To develop strategies for improving LLM response generation, focusing on contextual refinement and strict alignment with the retrieved content to minimize hallucinations.
- 4) **Comprehensive Evaluation:** To evaluate Qubo's reliability, accuracy, and usability through system testing and student assessments, determining its potential as a scalable and effective supplemental learning tool.

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II. RELATED WORK

A. Artificial Intelligence in Education

Artificial Intelligence (AI) has gradually become common in educational environments. Initially, AI applications were limited to automating tasks such as checking quizzes, recording grades, and arranging class schedules [12], [13]. These functions helped teachers save time and reduce repetitive workloads. Later, AI began to play a more direct role in learning, assisting students with review, explanation, and tutoring through Large Language Models (LLMs) that can adapt to learners' individual needs [14]. Beyond tutoring, AI systems have been designed to recommend learning materials, provide feedback, and adapt instructional content, transforming AI into a more interactive learning partner [15].

However, overreliance on AI poses several concerns. Excessive dependence can limit students' development of critical thinking and problem-solving skills [16]. Furthermore, AI-generated responses can sometimes be inaccurate or overly simplistic, misleading learners who lack the means to verify information [17]. Ethical issues such as data privacy, bias in model outputs, and unequal access to AI tools remain significant challenges [18]. Many scholars agree that AI should complement, not replace, the human educator's role in enriching learning experiences [19].

B. Chatbots with Large Language Models

Chatbots are among the most common AI-driven tools in education, capable of answering questions, providing hints, and explaining lessons [16]. Their accessibility and user-friendly nature make them valuable aids for self-study and academic support. However, the quality of chatbot responses varies widely. Some provide only surface-level or incomplete answers, particularly for complex or multi-step problems [19], [20]. Others may generate responses that appear accurate but are factually incorrect or misleading, a phenomenon known as hallucination [21], [22]. Consequently, ensuring the reliability and factual accuracy of chatbot responses remains an important research priority.

C. The Problem of Hallucination

Hallucination refers to the generation of plausible but incorrect information by AI systems [23]. This problem arises because LLMs are trained on static datasets and cannot always

update their knowledge with the latest information [24]. In educational contexts, hallucination can lead students to accept false or misleading answers, undermining their trust in AI systems. Ji *et al.* [25] emphasized that hallucination not only spreads misinformation but also diminishes AI's credibility as an educational tool. Researchers have observed that hallucinations often increase when AI models are prompted on unfamiliar or out-of-domain topics, reinforcing the need for grounding mechanisms and reliable verification methods [26].

D. Retrieval-Augmented Generation (RAG)

To address hallucination, researchers have proposed Retrieval-Augmented Generation (RAG). In this approach, the model first retrieves relevant information from external knowledge sources before generating responses [27]. By grounding answers in factual and up-to-date references, RAG reduces dependence on internal training data and enhances response accuracy [28], [29]. Despite its improvements over standalone LLMs, RAG still faces limitations. If the retrieval component fails to fetch relevant sources, the generated response may remain inaccurate or incoherent. Balancing factual accuracy and natural language fluency continues to be an open challenge [30].

E. Developments in RAG

Recent advancements have focused on strengthening RAG through enhanced architectures and fine-tuning strategies. Some systems incorporate hallucination detection modules that filter out fabricated content during generation [31]. Others emphasize transparency by tracing which documents influence final outputs or by visualizing reasoning paths for explainability [31]. State-of-the-art RAG models also integrate semantic search, re-ranking, caching, and feedback loops to improve both efficiency and retrieval precision [32]. Studies reveal a clear progression: early RAG systems (2017–2019) used basic retrieval techniques, while modern implementations (2024–2025) now include multimodal, adaptive, and domain-specific capabilities [33].

F. RAG in Education

RAG has recently been explored in educational contexts. Husain *et al.* [34] developed a RAG-based academic chatbot capable of providing accurate responses while filtering irrelevant queries. Similarly, Soliman *et al.* [35] introduced the BiWi AI Tutor, which integrates RAG with academic course materials, achieving up to 87% accuracy in question-answering tasks. Another system, MARK, designed for engineering education, combined keyword, sparse, and dense retrieval with reranking to ground answers in course-specific content [36]. While results are promising, RAG systems still face challenges related to limited database coverage, domain generalization, and personalization for diverse learners.

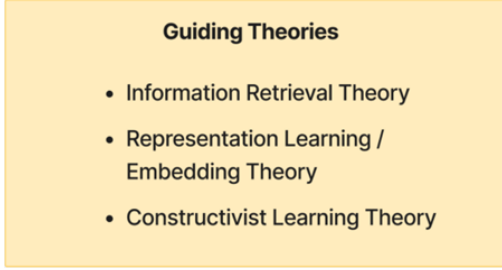
G. Synthesis

The reviewed literature highlights three core insights. First, AI and LLM-based chatbots increasingly shape how students

learn by providing quick, adaptable support. Second, these systems still exhibit critical weaknesses, including hallucination and limited personalization. Third, RAG provides a viable solution by integrating retrieval mechanisms that ground responses in authoritative sources. Nevertheless, RAG must be carefully designed, evaluated, and validated before it can be fully trusted as an educational technology.

III. METHODOLOGY

A. Guiding Theories



This study is grounded on the convergence of information retrieval, representation learning, and constructivist learning theory. Information retrieval provides the foundation for locating relevant text passages from lecture materials, while representation learning enables semantic encoding of these texts into high-dimensional vector embeddings. Constructivism emphasizes that students actively construct knowledge through meaningful engagement, aligning with the system’s goal of providing contextually faithful answers grounded in course materials.

B. System Overview



Fig. 1. System Overview Diagram

This study presents the design of Qubo using an Input–Process–Output (IPO) model. The input includes user queries and uploaded lecture notes. The process involves indexing, retrieval, reranking, and generation using a Retrieval-Augmented Generation (RAG) pipeline. The output provides accurate, reference-based responses that enhance students’ understanding of their course materials.

C. Large Language Model

The Mistral-Instruct large language model (LLM) serves as the generator for Qubo’s RAG layer. This open-source model is fine-tuned for instruction-following tasks, enabling context-aware, concise, and accurate responses. Its efficiency allows strong performance without requiring additional fine-tuning or training.

D. Prompt Assembly and Engineering

```

prompt = """
If context is empty or irrelevant say "I cannot answer that because I do not have information about it".

Context:
{context}

Question:
{query}

Answer:
"""
  
```

Prompt assembly involves integrating the user’s query, the retrieved documents, and system instructions into a single structured input. The prompt template explicitly instructs the model to respond only when sufficient contextual evidence is provided, reducing hallucinations and ensuring factual grounding. To address the “middle curse” problem—where LLMs tend to overlook information in the middle of a long prompt—a windowing strategy was applied, positioning the most relevant context at the beginning and end of the input.

E. System Architecture

Qubo’s architecture is composed of three primary layers: Frontend Layer – Built using React.js, it provides the chat interface, document upload manager, and user authentication. JWT (JSON Web Tokens) ensure secure access, while HTTPS encryption protects data transfer. Backend Layer – Developed with FastAPI, it manages API requests, handles authentication, and processes retrieval and generation workflows asynchronously using Uvicorn for high-speed performance. RAG Layer – The system’s intelligence core that integrates retrieval and generation. It retrieves relevant lecture note passages using FAISS and reorders them via BAAI/bge-reranker-base. The final answer is synthesized by Mistral-Instruct, producing factually consistent, context-grounded responses.

F. Retrieval and Similarity Computation

The retrieval stage combines lexical and semantic search to improve contextual relevance.



BM25 ranks text segments using term frequency and inverse document frequency, quantifying the importance of query terms within each document.

$$BM25(D, Q) = \sum_{t \in Q} IDF(t) \cdot \frac{f(t, D) \cdot (k_1 + 1)}{f(t, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{avgdl}\right)}$$

$$IDF(t) = \ln \left(\frac{N - n_t + 0.5}{n_t + 0.5} + 1 \right)$$

Vector embeddings are generated through MiniLM and stored in FAISS for efficient similarity search. Cosine similarity determines the semantic closeness between the

query and document embeddings.

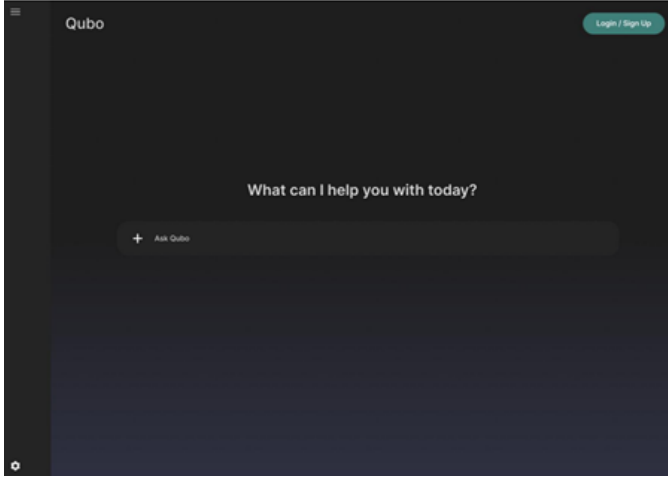
$$\text{CosineSimilarity}(A, B) = \frac{A \cdot B}{|A| |B|}$$

A fusion scoring method integrates BM25 and cosine similarity results, balancing lexical and semantic relevance for improved retrieval precision.

$$\text{FusedScore}(d) = \alpha \cdot \text{sdense}(d) + (1 - \alpha) \cdot \text{sBM25}(d),$$

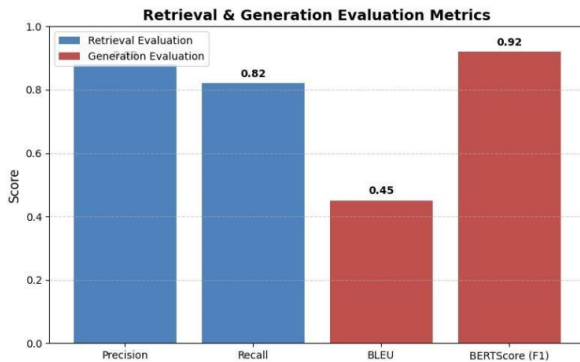
G. Frontend and Backend Integration

The frontend communicates with the backend through secure REST APIs. Students log in, upload course materials, and interact with the chatbot in real time. Uploaded documents are automatically indexed, embedded, and stored in the knowledge base. Queries are then routed through the RAG layer, and generated answers are displayed within the chat interface.



This modular integration ensures scalability, responsiveness, and security for multiple concurrent users.

IV. RESULTS



The system's ability to retrieve relevant documents was assessed using precision and recall. The results demonstrate a strong capability to identify pertinent information from the knowledge base.

- **Precision:** The system achieved a precision score of 0.88. This indicates that 88% of the documents retrieved by the

system were correctly identified as relevant to the user's query.

- **Recall:** The system achieved a recall score of 0.82. This means the system successfully found and retrieved 82% of all the relevant documents available in the knowledge base for a given query.

These metrics suggest that the retrieval component is effective, with a high rate of relevant document retrieval and a low rate of irrelevant document inclusion.

A. Generation Quality Evaluation

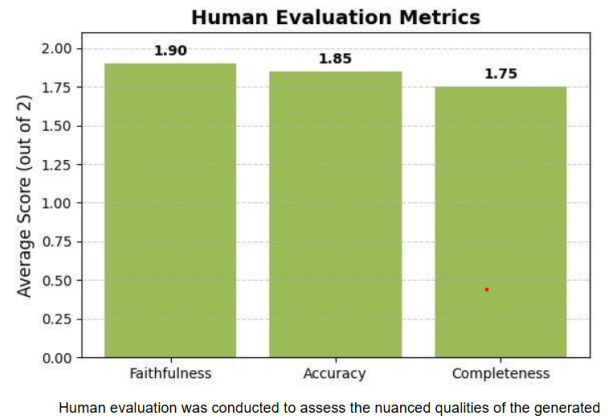
The quality of the generated responses was assessed using both automatic metrics and a rigorous human evaluation process.

B. Automatic Evaluation

The automatic metrics provided a quantitative measure of the semantic and syntactic similarity between the generated responses and the ground-truth answers.

- **BLEU Score:** The system obtained an average BLEU score of 0.45, indicating a moderate level of lexical overlap and grammatical similarity with the reference answers.
- **BERTScore:** The system achieved an average BERTScore (F1) of 0.92, demonstrating a high degree of semantic similarity and contextual alignment between the generated text and the reference answers.

C. Human Evaluation



Human evaluation was conducted to assess the nuanced qualities of the generated answers based on faithfulness, accuracy, and completeness, using a predefined 2-point scale. The average scores across all 300 queries are summarized in Table I.

Overall, the human evaluation confirms that the RAG system produces high-quality answers that are factually sound, faithful to the source material, and largely comprehensive.

TABLE I
HUMAN EVALUATION OF GENERATED ANSWERS

Criterion	Average Score (out of 2)	Interpretation
Faithfulness	1.90	Responses were highly consistent with the source lecture notes, with very few hallucinations or unsupported claims.
Accuracy	1.85	The vast majority of responses were factually correct and aligned with the information in the provided documents.
Completeness	1.75	Responses generally addressed the queries, though some occasionally omitted details.

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